

These problems should be completed by hand. Show your work and clearly label your final answers. You may use a calculator for arithmetic, but you should not use Python or another programming language.

1 Design matrices and dimensions

A housing dataset contains two features for each observation: size (in hundreds of square feet) and age (in decades). The response is price (in hundreds of thousands of dollars).

Observation	Size, x_1	Age, x_2	Price, y
1	12	2	3
2	18	1	5
3	15	4	4

(a) Write the feature matrix $\mathbf{X}$ without an intercept column and the response vector $\mathbf{y}$. State the shape of each.

(b) Write the augmented design matrix $\tilde{\mathbf{X}}$ after adding an intercept column of ones. Then write a compatible weight vector $\mathbf{w}$ using the symbols w_0 , w_1 , and w_2 .

(c) State the dimensions of $\tilde{\mathbf{X}}$, $\tilde{\mathbf{X}}^\top$, $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$, $\mathbf{w}$, $\tilde{\mathbf{X}}\mathbf{w}$, and $\tilde{\mathbf{X}}^\top\mathbf{y}$.

(d) Circle each valid product below. For every valid product, give the shape of the result.

$$\tilde{\mathbf{X}}\mathbf{w} \quad \mathbf{w}\tilde{\mathbf{X}} \quad \tilde{\mathbf{X}}^\top\mathbf{y} \quad \tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top \quad \tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}.$$

2 Predictions, residuals, and regression cost

Suppose the intercept column has already been included:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The regression cost used in this homework is

$$J(\mathbf{w}) = \frac{1}{2} \sum_i (\hat{y}_i - y_i)^2.$$

- (a) Compute the prediction vector $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$. Show each row-vector dot product.

- (b) Compute the residual vector $\hat{\mathbf{y}} - \mathbf{y}$. For each observation, indicate whether the model overpredicts, underpredicts, or predicts exactly.

- (c) Compute $J(\mathbf{w})$. Show the squaring and summation before giving the final value.

- (d) Now let $\mathbf{w} = [0 \ 1]^\top$. Calculate the new predictions and cost. Which weight vector is better for these two examples? Explain using the cost.

3 Stochastic gradient descent by hand

We have one feature and two training examples: $(x_1, y_1) = (3, 0)$ and $(x_2, y_2) = (7, 1)$. The model is

$$\hat{y} = w_0 + w_1x.$$

Initialize $w_0 = 0$ and $w_1 = 0$, use $\alpha = 0.1$, and visit example 2 before example 1. For one example, use $\tilde{\mathbf{x}}_i = [1 \ x_i]^\top$ and

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha(\mathbf{w} \cdot \tilde{\mathbf{x}}_i - y_i)\tilde{\mathbf{x}}_i.$$

(a) Using example 2, calculate the prediction, error (prediction minus response), update vector, and new values of w_0 and w_1 .

(b) Using the updated weights and example 1, repeat all four calculations.

(c) On the axes below, plot both observations and draw three lines: the initial model, the model after the first update, and the model after the second update. Label each line.



4 The analytic least-squares solution

For two observations, the augmented design matrix and response vector are

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Use

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

To keep the arithmetic focused, you may use

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{4} \begin{bmatrix} 10 & -4 \\ -4 & 2 \end{bmatrix}.$$

(a) Compute $\mathbf{X}^\top \mathbf{y}$.

(b) Multiply $(\mathbf{X}^\top \mathbf{X})^{-1}$ by your answer to find $\hat{\mathbf{w}}$. Clearly identify the intercept and slope.

(c) Write the fitted line $\hat{y} = \hat{w}_0 + \hat{w}_1 x$. Use it to predict y for $x = 1$ and $x = 3$.

(d) Calculate both residuals and $J(\hat{\mathbf{w}})$. What does the cost tell you about the fit?

(e) The homework code uses a pseudoinverse rather than an ordinary inverse. In one or two sentences, explain why that is useful.

5 Normalization by hand

A feature has values

$$\mathbf{x} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}.$$

Its mean is 20 and its standard deviation, using NumPy's default population formula, is approximately 8.165. Use

$$z_i = \frac{x_i - \text{mean}(\mathbf{x})}{\text{std}(\mathbf{x})}.$$

- (a) Normalize all three observations. Round to three decimal places.

- (b) Calculate the mean of your three normalized values. What do you notice?

- (c) Which observation is below the mean, at the mean, and above the mean? Explain how the signs of the normalized values communicate this.

- (d) Suppose a second feature is measured in hundreds of thousands. Explain why using the two raw features can cause difficulty for gradient descent and why normalization helps.

- (e) Why should the intercept column of ones not be normalized? What problem would occur if you attempted it?

6 Interpreting fitted weights

Suppose both the response (house price) and all features have been normalized. A fitted model is

$$\hat{y} = 0.45(\text{income}) - 0.08(\text{age}) + 0.30(\text{rooms}) + 0.02(\text{population}).$$

- (a) Which feature has the largest estimated effect in magnitude? Which has the smallest?

- (b) Interpret the coefficient 0.45 for income. Your answer should use standard deviations because both the feature and response are normalized.

- (c) Interpret the negative sign on the age coefficient.

- (d) Why is it more meaningful to compare these coefficients after normalization than before normalization?

- (e) Can this model establish that increasing income *causes* house prices to increase? Explain briefly.

- (f) The intercept is omitted from the displayed equation. If all features and the response are centered at zero, what intercept would you expect approximately? Why?

7 Matrix multiplication and runtime

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

(a) Compute $\mathbf{AB}$. In each position of the result, first write the corresponding row-column dot product, such as (row 1 of $\mathbf{A}$) $\cdot$ (column 1 of $\mathbf{B}$).

(b) Compute $\mathbf{BA}$. Is $\mathbf{BA}$ equal to $\mathbf{AB}$? What does this demonstrate about matrix multiplication?

(c) In general, $\mathbf{A}$ has shape (m, n) and $\mathbf{B}$ has shape (p, q) . What condition must hold for $\mathbf{AB}$ to exist, and what is the shape of $\mathbf{AB}$?

(d) A from-scratch implementation uses three nested loops: over rows of $\mathbf{A}$, columns of $\mathbf{B}$, and entries in each dot product. Give its runtime in terms of m , n , and q . Then simplify it when all three dimensions are approximately N .

(e) In one sentence, explain why this runtime helps motivate alternatives to repeatedly computing a closed-form least-squares solution on very large datasets.